



COCKRELL SCHOOL OF ENGINEERING
McKetta Department of Chemical Engineering

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ChE 350: Chemical Engineering Materials

SPRING 2026 #16515

Instructor: Dr. Kent Zheng

Office hour: Wednesdays 5:00-6:00 pm @ CPE 3.404

Email: kentz@utexas.edu

CLASS TIMES: 11:00 am – noon on Monday, Wednesday, and Friday in CPE 2.216.

RECITATION TIMES: 8:00 am – 9:00 am on Friday in CPE 2.216.

PREREQUISITES: Ch 353 with a grade of at least a C-.

COURSE DESCRIPTION: This course provides an introduction to key topics in contemporary materials science and engineering. Materials science focuses on understanding the structure-property relationships of solid phases; i.e., how materials respond to applied stimuli, such as light, electricity, and mechanical forces. By understanding the fundamental relationships between microscopic structure and properties, materials can be designed and fabricated to elicit a desired set of properties.

OVERALL COURSE ORGANIZATION: The course is organized in three major parts: 1) Materials Structures, 2) Materials Properties, and 3) Materials Synthesis: Thermodynamics and Kinetics.

COURSE MATERIALS:

- Lecture Notes (slides);
- Textbook: Brian S. Mitchell, "**An Introduction to Materials Engineering and Science for Chemical and Materials Engineers**," Wiley, New Jersey, 2004.
- Selected readings from supplementary books.
 - Allen & Thomas, "*The Structure of Materials*," Wiley, 1999.
 - Lovett, "*Tensor Properties of Crystals*," CRC Press, 2018.
 - Kittel, "*Introduction to Solid State Physics*," Wiley, 2005.
 - Porter & Easterling, "*Phase Transformations in Metals and Alloys*," CRC press, 2009.

COURSE REQUIREMENTS: The coursework consists of lectures, recitation sessions, readings, homeworks, quizzes, three mid-term exams, and the final exam.

GRADING:

Homeworks (weekly, due Friday, beginning of class)	10%
Quizzes (weekly, in the middle of each recitation)	5%
Mid-term exams (9/19, 10/24, and 11/21; lecture)	60%
Final exam (12/11, 3:30-5:30 pm)	25%

GRADES ARE NOT CURVED: >88% (A); 85-88 (A-); 82-85 (B+); 73-82% (B); 70-73 (B-); 67-70 (C+); 58-67% (C); 55-58 (C-); 50-55% (D); <50% (F)

APPENDIX I. TOPICS IN MATERIALS SCIENCE AND ENGINEERING*

	Class Topic
Structure	Introduction: A Roadmap for Materials Innovation
	Atomic structure and interatomic interaction
	Crystals: notation and symmetry
	Characterization of crystals: diffraction and microscopy
	Defects in crystals: 0D, 1D, 2D, 3D
	Glass and polymer
Properties	Mechanical properties: alloy, ceramic, and composite
	Electronic properties: metal, insulator, semiconductor, and devices
	Electrochemical properties: corrosion, batteries, and catalysis
	Case study: Nanomaterials
Synthesis	Phase equilibria: thermodynamics of solids, and phase diagram
	Diffusion in solids
	Nucleation and growth
	Diffusional and diffusion-less phase transformation in solids
	Case study: Dendritic growth in batteries
	Final exam (comprehensive)

* SUBJECT TO CHANGE.